Stat 441 Winter 2026
Instructor: Wenlu TANG
Assignment 1
Due date: Jan 26, 2026 before 6:00 pm

There are totally eight questions. Please use either words or latex or jupyternotebook to write your assignment. Include all necessary Python output in your answers. Please submit the **pdf** *files to the Blackboard system. Please* **DO NOT** *submit your Python code. Please show your courtesy to them by* $\boxed{\textit{boxing}}$ *your final solutions if that is not difficult, like*

$$\textit{...therefore the desired probability is } \boxed{P(A) = 0.334}.$$

In all the computations, if you need to round a number, please keep four decimal places, like

$$\frac{1000}{7} = 142.8571, \qquad \frac{0.02}{7} = 0.0029.$$

1. (5 marks) Prove that in simple linear regression,

$$\begin{aligned} \mathrm{R}^2 &= \frac{\mathrm{TSS} - \mathrm{RSS}}{\mathrm{TSS}} = 1 - \frac{\mathrm{RSS}}{\mathrm{TSS}} \\ &= \mathrm{corr}^2(x, y) \\ &= \left[\frac{\sum_i (y_i - \bar{y})(x_i - \bar{x})}{\sqrt{\sum_i (y_i - \bar{y})^2} \sqrt{\sum_i (x_i - \bar{x})^2}} \right]^2 \end{aligned}$$

2. I collect a set of data ($n = 100$ observations) containing a single predictor and a quantitative response. I then fit a linear regression model to the data, as well as a separate cubic regression, i.e. $Y = \beta_0 + \beta_1 X + \beta_2 X^2 + \beta_3 X^3 + \epsilon$.

   (a) (5 marks) Suppose that the true relationship between X and Y is linear, i.e. $Y = \beta_0 + \beta_1 X + \epsilon$. Consider the training residual sum of squares (RSS) for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

   (b) (5 marks) Answer (a) using test rather than training RSS.

   (c) (5 marks) Suppose that the true relationship between X and Y is not linear, but we don't know how far it is from linear. Consider the training RSS for the linear regression, and also the training RSS for the cubic regression. Would we expect one to be lower than the other, would we expect them to be the same, or is there not enough information to tell? Justify your answer.

   (d) (5 marks) Answer (c) using test rather than training RSS.

3. It is well-known that ridge regression tends to give similar coefficient values to correlated variables, whereas the lasso may give quite different coefficient values to correlated variables. We will now explore this property in a very simple setting.

   Suppose that $n = 2, p = 2, x_{11} = x_{12}, x_{21} = x_{22}$. Furthermore, suppose that $y_1 + y_2 = 0$ and $x_{11} + x_{21} = 0$ and $x_{12} + x_{22} = 0$, so that the estimate for the intercept in a least squares, ridge regression, or lasso model is zero: $\hat{\beta}_0 = 0$.

   (a) (5 marks) Write out the ridge regression optimization problem in this setting.

   (b) (5 marks) Argue that in this setting, the ridge coefficient estimates satisfy $\hat{\beta}_1 = \hat{\beta}_2$.

   (c) (5 marks) Write out the lasso optimization problem in this setting.

   (d) (5 marks) Argue that in this setting, the lasso coefficients $\hat{\beta}_1$ and $\hat{\beta}_2$ are not unique - in other words, there are many possible solutions to the optimization problem in (c). Describe these solutions.

4. Let $\lambda > 0$. Define

$$x^*(\lambda) = \arg\min_{x \in \mathbb{R}} \left\{ (x-1)^2 + \lambda |x| \right\}.$$

   That is, $x^*(\lambda)$ is the number such that for any $x \in \mathbb{R}$, $(x^*(\lambda)-1)^2+\lambda|x^*(\lambda)| \leq (x-1)^2+\lambda|x|$.

   (a) (5 marks) Prove that for $x \in \mathbb{R}$, if $x \neq x^*(\lambda)$, then $(x^*(\lambda) - 1)^2 + \lambda|x^*(\lambda)| < (x-1)^2 + \lambda|x|$.

   (b) (5 marks) Find $x^*(\lambda)$. Find all the λ such that $x^*(\lambda) = 0$.

5. (10 marks) Show that $\left\| \hat{\beta}^{\text{ridge}} \right\|$ increases as its tuning parameter $\lambda \to 0$. Does the same property hold for the Lasso estimator?

6. Let $X = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$ with x_1 and x_2 independently drawn from the standard normal distribution $N(0,1)$. Let $Y = A(X + \mu)$ with A being a 2×2 real matrix and $\mu \in \mathbb{R}^2$. Assume that $Y \sim N(\mu_0, \Sigma)$ with

$$\mu_0 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad \text{and} \quad \Sigma = \begin{pmatrix} \frac{5}{4} & -\frac{\sqrt{3}}{4} \\ -\frac{\sqrt{3}}{4} & \frac{7}{4} \end{pmatrix}.$$

   For your reference, a singular value decomposition of Σ is $\Sigma = UDU'$ with

$$U = \begin{pmatrix} \frac{1}{2} & \frac{\sqrt{3}}{2} \\ -\frac{\sqrt{3}}{2} & \frac{1}{2} \end{pmatrix}, \quad \text{and} \quad D = \text{diag}\{2, 1\}.$$

   (a) Find A and μ. (5 marks)

   (b) Find the distribution of $X + Y$. (5 marks)

7. The following code is made to draw independent random vectors from a two-dimensional normal distribution $N(\mu, \Sigma)$, with

$$\mu = \begin{pmatrix} -2 \\ 3 \end{pmatrix}, \quad \text{and} \quad \Sigma = \begin{pmatrix} 5 & 11 \\ 11 & 25 \end{pmatrix}.$$

Suppose one has a sample $x_1, \ldots, x_n \in \mathbb{R}^2$ drawn independently from $N(\mu, \Sigma)$. For each $1 \le i \le n$, denote

$$x_i = \begin{pmatrix} x_i^1 \\ x_i^2 \end{pmatrix}.$$

One may use the following estimators to verify the code

$$\hat{\mu} = \frac{1}{n} \sum_{i=1}^{n} x_i, \quad \text{and} \quad \hat{\Sigma} = \frac{1}{n-1} \sum_{i=1}^{n} (x_i - \hat{\mu})(x_i - \hat{\mu})^T. \tag{1}$$

It is expected that for large n, $\hat{\mu} \approx \mu$ and $\hat{\Sigma} \approx \Sigma$.

```
import numpy as np
np.random.seed(123)
def genData(n):
    mu = np.array([-2.0, 3.0])
    A = np.array([[1.0, 2.0], [3.0, 4.0]])
    X = np.random.normal(size = n * 2)
    X = X.reshape((2, n))
    X = (X + mu).dot(A)
    return(X.T)
```

(a) (5 marks) The current version of the function `genData` does not work properly. Please find out the problem and fix it (you may input the function into Python3, run it, and read the error message for a hint, but also keep in mind that there might be a problem in the code even if it runs smoothly). Please minimize your change of the code, so as to minimize the workload of grading. Please do not change the output format: the function `genData` outputs a matrix `X` of dimension `2×n`, with

$$\texttt{X}[j-1, i-1] = x_i^j, \quad 1 \le j \le 2, \quad 1 \le i \le n.$$

Please describe how you fix the problem (e.g. "change the code line ... to ...").

(b) (5 marks) After fixing the problem of `genData`, please run it with $n = 10$, and copy x_1 and x_2 to your solutions. Please set the random seed `123` before running `genData` so as to be consistent with the model solutions.

(c) (5 marks) Please reset the random seed `123` and run `genData` with $n = 10000$. Please use Formula (1) to compute $\hat{\mu}$ and $\hat{\Sigma}$. Please copy $\hat{\mu}$ and $\hat{\Sigma}$ to your solutions.

(d) (10 marks) Set the random seed `123` and generate 1000 data points $\{x_i\}_{i=1}^{1000}$ with the function `genData`. Then, make the 1000×1000 Gram matrix $\mathbb{K}$ with the kernel

$$K(x, u) = \exp(-\|x - u\|^2).$$

Find the largest three eigenvalues of $\mathbb{K}$. Here we use the Euclidean norm $\|\xi\| := \sqrt{\xi_1^2 + \xi_2^2}$ for $\xi = (\xi_1, \xi_2)^T \in \mathbb{R}^2$. Instead of using `numpy.linalg.eig`, you may use the function `numpy.linalg.eigh` for computing eigenvalues.